Ex:- $T_1 : \mathbb{R}^3 \to \mathbb{R}^2$

$T_2 : \mathbb{R}^3 \to \mathbb{R}^2$

$T_1((x,y,z)) = (x+y, y+z)$

$T_2((x,y,z)) = (x, y+z)$

$(T_1 + T_2)(x,y,z) = T_1(x,y,z) + T_2(x,y,z)$

$= (x+y, y+z) + (x, y+z)$

$= (2x+y, 2y+2z)$

* $T : U \to V$

$S : V \to W$

$S \circ T = ST(u)$

$= S(v)$

$= w.$

Nilpotent operator:-

A linear transformation T on a v.s V is said to be nilpotent if $T^n = 0$, where n is a least positive integer where n is known as the nilpotency index of T.

Ex:- Differential operator

$D : P_n \to P_n$

$D^{n+1} = 0$

2) $T : \mathbb{R}^3 \to \mathbb{R}^3$

$T(x,y,z) = (0,x,y)$

$T^2(x,y,z) = T(0,x,y) = (0,0,x)$

$T^3(x,y,z) = T(0,0,x) = (0,0,0).$

$\therefore$ nilpotency index of $T = 3$

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